

Diabetic Retinopathy Classification Using Deep Learning Techniques



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ABSTRACT

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LIST OF ABBREVIATIONS

DL	Deep Learning
DR	Diabetic Retinopathy
NPDR	Non-proliferative diabetic retinopathy (NPDR)
PDR	Proliferative diabetic retinopathy
CNN	Convolutional Neural Network
ROI	Regions of Interest

LIST OF SYMBOLS

Chapter 1

Introduction

1.1 Introduction

The human eye is an amazing sensory organ that is the gateway to visual perception. It detects light and sends signals to the brain through the optic nerve to form images, colors and depth perception [1]. This complex process allows us to recognize face's help detect objects and do many activities that require visual input. But many medical conditions can impair the eye and cause significant visual loss, Diabetic Retinopathy (DR) is one of them.

Diabetic retinopathy is a complication of diabetes that leads to vision loss. It happens due to prolonged diabetes that damage blood vessels in the retina and is the prominent reason for blindness worldwide. It affects the thin blood vessels in the retina that are responsible for capturing light and can cause blockage and bleeding in the retina, and ultimately leads to permanent vision loss if not detected early. Roughly, one-third of the 463 million people worldwide who have diabetes are affected by it [2]. The International Diabetes Federation (IDF) projects that there will be a significant increase in the global diabetes population from roughly 552 million in 2035 to 642 million in 2040 [3]. DR can be classified into five distinct stages including No DR (DR0), Mild (DR1), Moderate (DR2), Severe Non-proliferative (DR3), and Proliferative (DR4) [4]. Non-proliferative diabetic retinopathy (NPDR) causes retinal swelling and leakage of tiny blood vessels while Proliferative diabetic retinopathy (PDR) is the most severe stage of the disease, in which new blood vessels start developing in the retina.

Deep learning (DL) methods have shown great promise in automatically detecting the accurate stages of DR. These models showed great accuracy in detecting DR but still many ophthalmologists heavily rely on manual detection. Manual detection of DR is time-consuming and requires trained clinical experts to analyze digital color fundus images. However, the delayed outcomes can result in a lack of follow-up and severe outcomes for patients [5]. Convolutional Neural Networks (CNNs) have showed great performance in recognizing DR-related features, such as microaneurysms, hemorrhages, and exudates, directly from images without the need for handcrafted features [paper mid analysis]. Despite the progress in machine learning based Diabetic Retinopathy (DR) detection, we need to address the current limitations. Existing models although

work well in many controlled settings, fail to generalize across different datasets and real-world clinical environments. This is mainly due to issues like: Data Imbalance, Limited diversity in training datasets, Subtle visual differences and inability to capture Regions of Interest (ROI) which are important for detection and classification of DR. Moreover, existing models have high computational cost that makes them unable to operate in resource limited areas.

Many DL techniques showed great accuracy in detecting and classifying DR but now the focus is towards using Hybrid techniques which combines different CNN models to make the output more clinically explainable because they don't only classify DR but also provide ophthalmologists enough information to use them in real time diagnosis [6]. Moreover, hybrid models also help's in detecting progression of disease.

This research will focus on advanced computational techniques, state of the art DL frameworks and robust validation processes for DR classification. Our model will integrate CNNs for local feature extraction with multi-stage attention mechanism for highlighting critical regions, and employs feature fusion to enhance the separation between classes.

The main goal of this research is to develop robust AI-based solution that is mostly required in both urban hospitals and rural healthcare centers [7]. This solution will not only reduce the workload of specialists but also make early classification more accurately. Moreover, with timely intervention, this technology could play a vital role in preventing vision loss and improving the quality of life for millions of diabetic patients worldwide.

1.2 Research Gaps

Existing DR research faces significant challenges due to limited dataset diversity, class imbalance, and the difficulty of distinguishing subtle differences between adjacent DR stages. Limited representation of variations in color, contrast, resolution, and imaging devices in training data results in lower accuracy and poor generalization of the models. Additionally, the unequal distribution of DR severity levels particularly the scarcity of advanced cases introduces bias and reduces reliable classification across all classes. These issues are further compounded by the subtle visual differences between early and intermediate DR stages, making precise classification.

1.3 Research Questions

RQ1: Can a hybrid deep learning model incorporating attention and feature fusion outperform standard CNN models in accurately classifying different stages of diabetic retinopathy?

RQ2: How does the integration of attention mechanisms and feature fusion help deep learning models handle image quality variations, class imbalance, and improve generalization across diverse real-world DR datasets?

1.4 Research Contributions

The contributions are summarized below:

1.5 FYP Document Organization

This FYP document is organized into five chapters. The chapter-wise description is given below:

Chapter 1 presents a detailed introduction regarding DR classification.

Chapter 2 gives a review of state-of-the-art techniques for DR classification

Chapter 3 describes the proposed methodology for DR classification.

Chapter 4 presents experimental results for proposed method. This chapter also includes a detailed discussion and comprehensive comparison with state-of-the-art studies.

Chapter 5 provides thesis conclusion, along with future research directions.

Chapter 2

Literature Review

2.1 Introduction

In the proceeding chapter, relevant literature is presented under different aspects of DR classification. The chapter include several studies mainly focusing on DL, hybrid and vision transformer techniques that systematically discuss state-of-the-art approaches for retinal lesion localization, attention-based feature extraction, and diabetic retinopathy severity grading.

2.2 Literature Review

Diabetes is one of the main health dilemmas worldwide. One complication of diabetes is diabetic retinopathy (DR), which is one of the main causes of vision loss worldwide. Manual DR detection using large fundus image data is time-consuming and error-prone. An effective automatic DR detection system can be significantly more efficient and potentially more accurate. This has motivated extensive research into automated DR classification using deep learning, particularly Convolutional Neural Networks (CNNs), which can learn discriminative visual features directly from retinal images without the need for handcrafted descriptors. Modern CNN-based systems have strong capability to detect clinically important lesions such as microaneurysms, hemorrhages, exudates, and vessel abnormalities, enabling reliable multi-stage DR grading suitable for real-world deployment. Traditional methods used handcrafted features with classical classifiers. They were simple and interpretable but sensitive to image quality and lacked scalability. Asian epidemiological studies using image enhancement and manual features achieved about 65-80% accuracy [8], while global DR screening studies using logistic regression and classical ML on Messidor and clinical datasets reported 70-80% performance [9].

Several studies in this domain have explored enhanced CNN pipelines to improve accuracy, robustness, and generalization across heterogeneous fundus datasets. A hybrid approach [10] combining blood vessel segmentation with Contrast Limited Adaptive Histogram Equalization (CLAHE) and threshold-based enhancement was proposed, where extracted features were fused and classified using VGG19 and InceptionV3 models. Evaluated on the Kaggle DR Detection dataset, this method achieved an accuracy of 96%, highlighting the effectiveness of lesion enhancement and vessel information in improving classification performance. In another study [11] a deep learning-based Retinopathy Severity Grading Network (RSG-NET) was developed for

multi-class DR grading. The model incorporated extensive preprocessing and data augmentation to improve generalization. When evaluated on the Messidor-1 dataset, the proposed CNN achieved a notably high accuracy of 99%, demonstrating the strength of customized CNN architectures for DR severity classification. To address computational efficiency, a lightweight deep learning model named RetNet-10 was introduced by optimizing a base CNN through careful hyperparameter tuning. This model was evaluated on a combined dataset consisting of APTOS, Messidor-2, and IDRiD images and achieved an accuracy of 98.65%, proving that high accuracy can be attained even with reduced model complexity [12]. A two-stage adaptive fine-tuned CNN approach was proposed for automatic DR diagnosis, integrating Principal Component Analysis (PCA) for feature reduction and Gradient Boosting for final classification. This framework was trained and evaluated on E-Ophtha, EyePACS, and Messidor datasets, achieving an accuracy of 95%. The study demonstrated that combining deep feature extraction with traditional machine learning classifiers can enhance diagnostic performance [13]. Another study [14] focused on early DR detection by implementing a five-layer CNN architecture consisting of Conv2D, max-pooling, and softmax layers. The model was trained on the APTOS dataset and tested on Messidor-2 and IDRiD datasets, achieving an accuracy of 95.27%, indicating robust cross-dataset generalization. Similarly, a lightweight CNN combined with CLAHE-based contrast enhancement, data augmentation, and Global Average Pooling (GAP) was proposed for early detection and severity classification of DR. Using the APTOS dataset, the method achieved an accuracy of 97.5%, emphasizing the role of contrast enhancement and architectural simplification in improving classification outcomes [15]. Most recently, efforts have been made to improve model interpretability by integrating Explainable Artificial Intelligence (XAI) techniques with deep learning models. A custom CNN architecture using convolutional, max-pooling, and dense layers with ReLU activation was developed and compared with transfer learning models on the APTOS dataset. The proposed approach achieved an accuracy of 95.27%, while also providing visual explanations to support clinical decision-making [16].

Hybrid deep learning approaches have emerged as a powerful alternative to single-model CNN architectures for Diabetic Retinopathy (DR) detection and classification. These methods integrate multiple deep learning models or combine deep feature extraction with traditional machine learning classifiers to address common challenges such as limited generalization, class imbalance, and lack of interpretability. Several

studies have demonstrated the effectiveness of CNN–CNN feature fusion. A hybrid CNN model [17] combining ResNet50 and InceptionV3 was proposed for automatic DR classification, where deep features extracted from both networks were fused before classification. Evaluated on the Kaggle DR dataset, this approach achieved an accuracy of 96.85%, outperforming individual backbone models by leveraging complementary spatial and contextual features. In another study [18] heterogeneous fundus data from multiple datasets (APTOS, Messidor-2, DDR, and IDRiD) were fused using a VGG19 and ResNet50V2 hybrid framework. Although the approach improved robustness across datasets, it achieved a comparatively lower accuracy of 91.82%, highlighting the challenges associated with dataset variability and domain shift in multi-source training. Lesion-focused hybrid models have also been explored to enhance sensitivity to pathological features. A lesion-based DR detection framework utilized a hybrid CNN architecture combining ResNet-16 and GoogLeNet, optimized using Adaptive Particle Swarm Optimization (APSO) and followed by traditional machine learning classifiers. Tested on the EyePACS dataset, this method achieved an accuracy of 94%, demonstrating the benefit of combining deep feature extraction with optimized classification strategies [19]. Beyond CNN fusion, sequential hybrid architectures have been proposed. A hybrid framework integrating CNNs with Recurrent Neural Networks (RNNs), specifically LSTM, was developed to model spatial and sequential feature dependencies. Feature fusion was employed to improve early DR detection on the DRIVE and EyePACS datasets however, the study did not report a specific accuracy, limiting direct performance comparison [20]. Attention-based hybrid models have further enhanced feature discrimination. A dual attention CNN incorporating spatial and channel attention mechanisms, combined with transformer encoder layers, was proposed to improve contextual understanding of retinal features. Evaluated on APTOS and EyePACS datasets, the model achieved an accuracy of 95.7%, demonstrating the effectiveness of attention-guided hybrid feature extraction [21]. Several studies have combined deep learning with classical machine learning classifiers to improve efficiency and interpretability. A transfer learning-based hybrid approach using GoogleNet and ResNet-18 for feature extraction, followed by an SVM classifier, achieved an accuracy of 89.29% on the APTOS dataset [22]. Similarly, a hybrid transfer learning framework [23] using NASNet-Large and ResNet-50, combined with green channel extraction (GCE) and CLAHE preprocessing, was proposed for early DR detection. Tested on the APTOS dataset, the method achieved an accuracy of 91.14%,

emphasizing the importance of color channel optimization and contrast enhancement in hybrid pipelines. Another study [24] combining EfficientNetV2 with an SVM classifier achieved an accuracy of 97.26% on the APTOS dataset, demonstrating that efficient architectures paired with traditional classifiers can yield competitive results.

Recent advancements in Diabetic Retinopathy (DR) classification have increasingly focused on attention mechanisms and Vision Transformers (ViTs) to overcome the limitations of conventional convolutional neural networks. Unlike CNNs, which primarily capture local spatial patterns through fixed receptive fields, transformer-based models process retinal images as a sequence of patches and apply self-attention to model long-range dependencies across the entire retina. A notable contribution in this domain is Hybrid Fusion-Net which integrates a Self-Attention Network (SAN) with a Vision Transformer (ViT) for multi-stage DR detection. By combining attention-driven global context modeling with deep feature learning, the framework effectively captures both local lesion details and global retinal structures. Evaluated on the Kaggle Diabetic Retinopathy Detection dataset, the model achieved a high accuracy of 99%, demonstrating the strong potential of transformer-based hybrid architectures for large-scale DR classification [25]. A CNN augmented with feature attention modules and transformer-inspired positional encoding emphasized lesion-relevant regions and achieved 95.70% accuracy on the APTOS dataset, highlighting the benefit of attention mechanisms without fully replacing CNNs [26]. Likewise the study by [27] a unified segmentation classification framework combining U-Net, a pretrained CNN, and a Vision Transformer was proposed to fuse pixel-level and global features. Tested on the Kaggle DR dataset, the model showed robust performance under challenging imaging conditions, though classification accuracy was not explicitly reported, limiting direct comparison.

2.3 Traditional Approaches

Traditional DR detection methods relied on handcrafted features with classical machine learning classifiers. They used image processing techniques such as contrast enhancement, morphological filtering, and edge detection to analyze retinal structures. Key features included blood vessel segmentation, optic disc localization, and lesion-specific textures like microaneurysms and hemorrhages. Classifiers such as SVM, Random Forest, k-NN, and Decision Trees were then used to grade DR severity. Although computationally efficient and interpretable, these methods were highly

sensitive to image quality and illumination variations. Due to poor scalability and heavy dependence on manual feature design, they have largely been replaced by deep learning-based approaches.

Table 2.1: Traditional Approaches for Diabetic Retinopathy

Ref	Title	Method/Approaches	Dataset	Acc (%)
[8]	Epidemiology of DR and macular edema in Asia: a systematic review and meta-analysis	Image Enhancement, Manual Feature Extraction	Asian hospital datasets	65–78%
[9]	DR: global prevalence, risk factors, screening practices and challenges	Screening Tools, Logistic Regression, Classical ML	Messidor, Duke, Clinical datasets	70–80%

2.4 Deep Learning Approaches

Deep learning especially Convolutional Neural Networks (CNNs) has transformed automated DR detection by eliminating the limits of manual, handcrafted features. CNNs learn layered features directly from retinal images, allowing them to detect both fine lesions (microaneurysms, hemorrhages, exudates) and overall retinal patterns. Popular models like VGG, Res-Net, Dense-Net, and Inception consistently outperform traditional methods in multi-class DR grading. These models work especially well on large datasets such as EyePACS and APTOS, where thousands of images help them learn strong, generalizable features. Approaches like transfer learning, fine-tuning, and data augmentation further boost accuracy when data is limited. CNN-based systems are often paired with preprocessing steps normalization, resizing, and lesion enhancement to reduce noise and highlight important structures. However, they still struggle with issues like class imbalance, reliance on annotated labels, and high training costs. Even with these challenges, CNNs remain the core of DR detection research because of their high accuracy, flexibility, and suitability for real-world clinical use.

Table 2.2: Deep Learning Approaches for Diabetic Retinopathy (2023-2025)

Ref	Title	Method/Approaches	Dataset	Acc (%)
[10]	DR Classification Using Hybrid Color-Based CLAHE and Blood Vessel in Deep CNN	BV segmentation + CLAHE-TH enhancement, fused features into VGG19 / InceptionV3 classifier.	Kaggle DR detection dataset	96%
[11]	A DL based model for diabetic retinopathy grading	Retinopathy Severity Grading Network (RSG-NET) CNN with preprocessing and augmentation.	Messidor-1	99%
[12]	A Lightweight Robust Deep Learning Model for DR classification	Improved base CNN through hyperparameter tuning to create RetNet-10.	APTOS, Messidor2, and IDRiD (combined)	98.65%
[13]	Automatic Diabetic Retinopathy Diagnosis Using Adaptive Fine-Tuned CNN	Two-stage CNN fine-tuning with PCA and Gradient Boosting.	E-optha(training), EyePACS, Messidor.	95%
[14]	Early Detection of Diabetic Retinopathy Through the Integration of Deep Learning Models	5-layer CNN with Conv2D, max-pooling and softmax for DR classification.	APTOS (training), Messidor2, IDRiD (testing)	95.27%
[15]	Early Detection and Severity Classification of Diabetic Retinopathy Using CNN	Lightweight CNN + CLAHE + Augmentation + GAP	APTOS	97.5%
[16]	Enhancing Early Detection of DR Through the Integration of Deep Learning Models and Explainable AI	Custom CNN with convolution, max-pooling, and dense layers using ReLU, compared against transfer learning models.	APTOS	95.27%

2.5 Hybrid Approaches

Hybrid models have gained attention as a strong alternative to using a single CNN because they merge different deep learning models. These approaches are designed to overcome issues like limited generalization, Lack of Interpretability and class imbalance. Typical hybrid models include combining networks such as Inception V3 with ResNet50, DenseNet with Inception blocks, VGG19 with ResNet50 or running parallel architectures that capture both low-level texture patterns and high-level semantic details. By combining complementary components, hybrid pipelines improve the depth and richness of features, increase model efficiency, and boost diagnostic performance across all DR grades. These systems handle varied image quality and subtle lesion differences more effectively. However, these benefits come with drawbacks hybrid methods usually require more computation, longer training, and can be harder to interpret. Even so, they provide a solid base for building scalable, multi-stage DR classification systems that often surpass single-

model setups.

Table 2.3: Hybrid Approaches for Diabetic Retinopathy (2023-2025)

Ref	Title	Method/Approaches	Dataset	Acc (%)
[17]	Hybrid CNN Model for Automatic DR Classification	combines ResNet50 and InceptionV3 features, fuses them, and classifies DR	Kaggle DR dataset	96.85%
[18]	Fusion of heterogeneous fundus data	VGG19 + ResNet50V2 / Hybrid Deep Learning	Aptos/Messidor-2/DDR/IDRID	91.82%
[19]	Lesion based DR detection using hybrid deep learning model	Hybrid CNN (ResNet-16 + Google-Net) with APSO + ML Classifiers	EyePACS	94%
[20]	Hybrid deep learning framework for early detection	CNN+RNN(LSTM)/Feature fusion	DRIVE/EyePACS	N/R
[21]	DR classification through dual attention mechanism	CNN with spatial and channel dual attention modules combined with trans former encoder layers for enhanced feature extraction	APTOS /EyePACS	95.7%
[22]	Diabetic Retinopathy Detection from Fundus Images of the Eye Using Hybrid DL Features	Transfer Learning-based Hybrid GoogleNet and ResNet-18 features with SVM Classifier	APTOS	89.29%
[23]	Early detection and classification of DR by transfer learning of NASNet-large and ResNet-50 CNN	Transfer learning with ResNet-50+ GCE (green channel extraction) + CLAHE	APTOS	91.14%
[24]	A Hybrid Model for Diabetic Retinopathy Classification Using Retinal Fundus Images	EfficientNetV2+SVM	APTOS	97.26%
[25]	HybridFusionNet: Deep Learning for Multi-Stage Diabetic Retinopathy Detection	hybrid deep learning approach combining Self-Attention Network (SAN)+Vision Transformer (Vit)	Kaggle Diabetic Retinopathy Detection dataset	99%

2.6 Vision Transformer Approaches

Recent progress in diabetic retinopathy (DR) classification includes the use of attention mechanisms and Vision Transformers (ViTs), which improve both accuracy and efficiency. These models can highlight clinically important retinal regions, such as

lesions and abnormal blood vessels, instead of processing all areas equally like conventional CNNs. Transformer-based approaches analyze images as patches and use self-attention to understand global relationships across the retina. When combined with CNN feature extractors, hybrid models capture fine local details along with long-range contextual information, leading to better performance under varying image quality and lighting conditions. Attention maps also make predictions more explainable. Although these methods demand high computational resources and large labeled datasets, and may struggle with noisy images, they represent a strong step toward scalable and clinically reliable DR detection systems.

Table 2.4: Vision Transformer Approaches for Diabetic Retinopathy (2023-2025)

Ref	Title	Method/Approaches	Dataset	Acc (%)
[25]	Hybrid Fusion-Net: Deep Learning for Multi-Stage Diabetic Retinopathy Detection	hybrid deep learning approach combining Self-Attention Network (SAN)+Vision Transformer (Vit)	Kaggle Diabetic Retinopathy Detection dataset	99%
[26]	Feature attention module on CNN for DR detection	CNN with feature attention modules and transformer-inspired positional encoding for highlighting lesion regions	APTOS	95.70%
[27]	DL powered segmentation and classification using transfer learning	Hybrid model combining U-Net segmentation, pretrained CNN, and Vision Transformer (ViT) for feature fusion and classification	Kaggle DR Dataset	

2.7 Survey Papers

Survey papers provide a comprehensive overview of existing diabetic retinopathy (DR) detection methods by systematically comparing datasets, model architectures, and performance metrics. They help identify research trends, strengths, limitations, and open challenges, thereby guiding future work in DR classification.

Table 2.5: Survey papers for comparison of studies (2023-2025)

Ref	Title	Method/Approaches	Dataset	Results
[28]	A Comprehensive Survey and Comparison of Methods	ResNet-50 + SVM	APTOS/EyePACS	Up-to 99% accuracy using hybrid DL methods
[29]	Diabetic Retinopathy Classification with Deep Learning via Fundus Images: survey	Multiple datasets	CNNs and RNNs	Deep learning models gained high accuracy
[30]	Deep Learning Techniques for Diabetic Retinopathy Classification: survey	Multiple Public datasets	deep learning techniques for Diabetic Retinopathy (DR)	transformer and self-supervised approaches give promising results

2.8 Public Fundus Datasets

Datasets are essential for training and evaluating deep learning (DL) models and have played a major role in recent advances in DL. However, collecting large and well-labeled fundus image datasets is difficult. One key challenge is patient privacy, which restricts access to and sharing of medical images. In addition, differences in fundus cameras, imaging conditions, and operators used, alters the image quality (such as size, noise, artifacts, contrast, illumination, and sharpened regions). Another challenge is manual labeling, which is usually done by ophthalmologists, and differences in their judgment can lead to inconsistent labels. To address these challenges, researchers often use publicly available fundus image datasets. Therefore, below table gives a concise overview of several publicly available datasets of fundus images for DR classification. Among public datasets, MESSIDOR [31], Kaggle EyePACS [32], Kaggle APTOS [33] and MESSIDOR-2 [34] are the most frequently used for DR classification. The Kaggle EyePACS dataset is the largest among them, containing 88,702 fundus images labeled according to the International Clinical Diabetic Retinopathy Severity Scale (ICDRSS) [35]. Images were collected using different cameras and under various imaging conditions, making the dataset suitable for real-world applications. Other datasets, including E-Ophtha [36], DRiDB [37], IDRID [38], and DDR [39], were created for both DR lesion detection and disease grading. The DRiDB dataset contains 50 fundus images with detailed labelling of the optic disc, blood vessels, DR lesions, and severity grades provided by multiple experts. Although it is small, it provides the largest amount of information and can be used for lesions detection and DR grading. The DDR dataset includes 12,522 fundus images labelled by several specialists following the ICDRSS

guidelines. However, the dataset shows a class imbalance across DR severity levels, thus, it has not been widely used. Since most of these datasets are small, many studies combine multiple datasets or use different datasets for training and validation of DL models to improve performance and generalization.

Table 2.6: Publicly Available DR Classification Datasets

Year	Dataset (Available link)	Country	Size	Annotations	Tasks
2004	MESSIDOR [31] http://www.adcis.net/en/third-party/messidor/	France	1200	Image level	DR grading
2009	Kaggle EyePACS [32] https://www.kaggle.com/c/diabetic-retinopathy-detection/data	United States	88702	Image level	DR grading
2010	MESSIDOR-2 [34] http://www.adcis.net/en/third-party/messidor2/	France	1748	Image level	DR grading
2013	E-Ophtha [36] http://www.adcis.net/en/third-party/e-ophtha/	France	463	Pixel level	DR diagnosis, Lesion detection
2013	DRiDB [37] https://ipg.fir.hr/ipg/resources/image-database#	Croatia	50	Pixel level	DR grading, Lesion detection
2017	IDRiD [38] https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid	India	516	Image and Pixel level	DR grading, Lesion detection
2019	Kaggle APTOS 2019 [33] https://www.kaggle.com/competitions/aptos2019-blindness-detection	India	3662	Image level	DR grading
2019	DDR [39] https://www.kaggle.com/datasets/mariaherrerot/ddrdataset	China	13673	Image and Pixel level	DR grading, Lesion detection

2.9 Summary

This chapter presents a comprehensive review of existing research on DR classification, highlighting recent advancements in DL, hybrid CNN and Vision Transformer models. The chosen studies mainly focus on feature extraction, feature selection, attention mechanisms, and classification strategies. Publicly available Benchmark datasets were used mostly, comparative results reported in the literature are also summarized. Based

on the limitations of prior work, research gaps are identified. These insights motivate the development of the proposed methodologies presented in Chapter 3.

Chapter 3

Proposed Methodology

3.1 Introduction

This chapter will present proposed method for DR classification using hybrid DL architecture. The prime concern of this methodology is to classify fundus normal and low-quality images with improved performance.

3.2 Proposed Methodology

This methodology consists of four steps such as data preparation, proposed hybrid DL architecture (CNN backbone, multi-stage attention block, feature fusion, and fully connected layer), classifier. The details related to these steps are given in the subsequent sections.

3.2.1 Data Preparation

Data Augmentation: Flips, rotations for class balancing and dataset diversity.

Super-Resolution Enhancement: As a preprocessing step we will apply SRGAN (Super-Resolution Generative Adversarial Network) to enhance low resolution fundus images, improving visibility.

3.2.2 Hybrid DL Architecture

CNN backbone: This step will be used for feature extraction. In this regard, we will apply VGG54 to extract features from fundus images.

Multi-stage Attention Block: This block refines the extracted features by focusing on important regions. It helps the model better distinguish different DR stages. Multistage attention module with four attention blocks is proposed to improve efficiency.

Cross-Patch Attention

Models relationships between different patches of the fundus image. Helps detect patterns spread across regions.

Global Context

Captures overall retinal structure and long-range dependencies. Useful for DR stage understanding.

Region Attention

Focuses on important retinal regions (lesions like microaneurysms, exudates).

Suppresses background.

Feature Attention Block

Reweights important feature channels (e.g., texture, color, vessels). Enhances discriminative features.

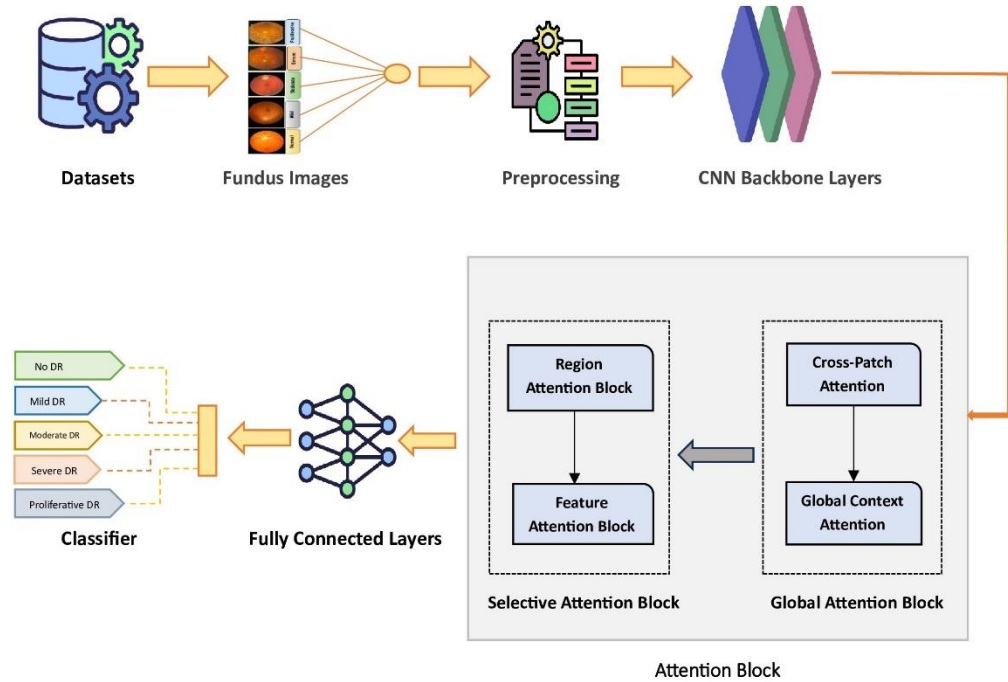


Figure 3.1: Flow Diagram of Proposed Methodology

Feature Fusion: A Combination of outputs from multiple layers of CNN.

Fully connected layers: The refined features are converted into a compact representation. This layer prepares the features for final classification.

3.2.3 Classifier:

The classifier predicts the diabetic retinopathy stage. The output includes No DR, Mild, Moderate, Severe, and Proliferative DR.

Chapter 4

Results and Discussions

4.1 Introduction

This chapter presents the experimental setup and results obtained after implementing proposed methodology for DR classification (as discussed in chapter 3). The evaluation of each task is described separately in subsequent sections.

4.2 Datasets Used

Table 4.1: Statistics of Applied Datasets

Datasets	Image type	No of images	Challenges
APTOS	Retinal Fundus Images	3,662 images	Small dataset size, Imbalance classes, Lesion Similarity
EyePACS	Retinal Fundus Images	88,702 images	Class imbalance, low resolution, Noise, illumination variations

4.3 Evaluation Metrics and Experimental Settings

Three evaluation metrics including Accuracy, specificity, Sensitivity are selected to estimate the performance of proposed method for DR classification. The experimental settings cover Adam (Adaptive Moment Estimation), learning rate schedule, activation functions, momentum, batch size, and epochs. We will apply K-fold as cross validation method. Moreover, Grad-CAM and lesion-specific saliency maps are used to highlight critical retinal regions influencing predictions.

4.4 Result Analysis

4.5 Comparison with Existing Studies

4.6 Summary

In this chapter, proposed methodology is evaluated on several publicly available challenging datasets.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

5.2 Future Work

Chapter 6

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